

Autodesk® Scaleform®

Scaleform

Scaleform 4.3 Scaleform

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3.02
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Autodesk® Scaleform® 4.3

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1

Autodesk® Scaleform® Video™ by CRI™ Scaleform™
Adobe® Flash® Flash
HUD

CRI CRI
CRI 1700

Scaleform PC (Xbox 360®)
PLAYSTATION®3”(PS3™) Wii™
Scaleform Alpha

Scaleform Adobe Premiere® Adobe After Effects®
Adobe Flash
5.1
Scaleform alpha

Scaleform Video SDK

- Scaleform
- Scaleform
- /
- Scaleform ()

2 **Scaleform**

Scaleform

2.1

Scaleform

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder

Scaleform

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video

Flash
videodemo.swf

USM

Scaleform

2.4

-

1080p API (HD)

-

-

SDK

2.5

Scaleform®
Scaleform

CRI Movie™

Adobe Flash®

3D

HUD

Scaleform
Adobe Flash

Scaleform
CRI Movie™

Scaleform™

2.6

- : AVI WAV
- : Scaleform Xbox 360 PS3 PC Wii
- :
- : Scaleform
- **Alpha** : alpha
“ ”
- : 3D
- : 32
- : using Scaleform 4.3 Flash ActionScript™
- : Scaleform ActionScript
- : UI
- : / 5.1
- :
- : Scaleform Video
- : Scaleform USM
Flash ActionScript

2.7

Scaleform

Scaleform CRI

USM

USM

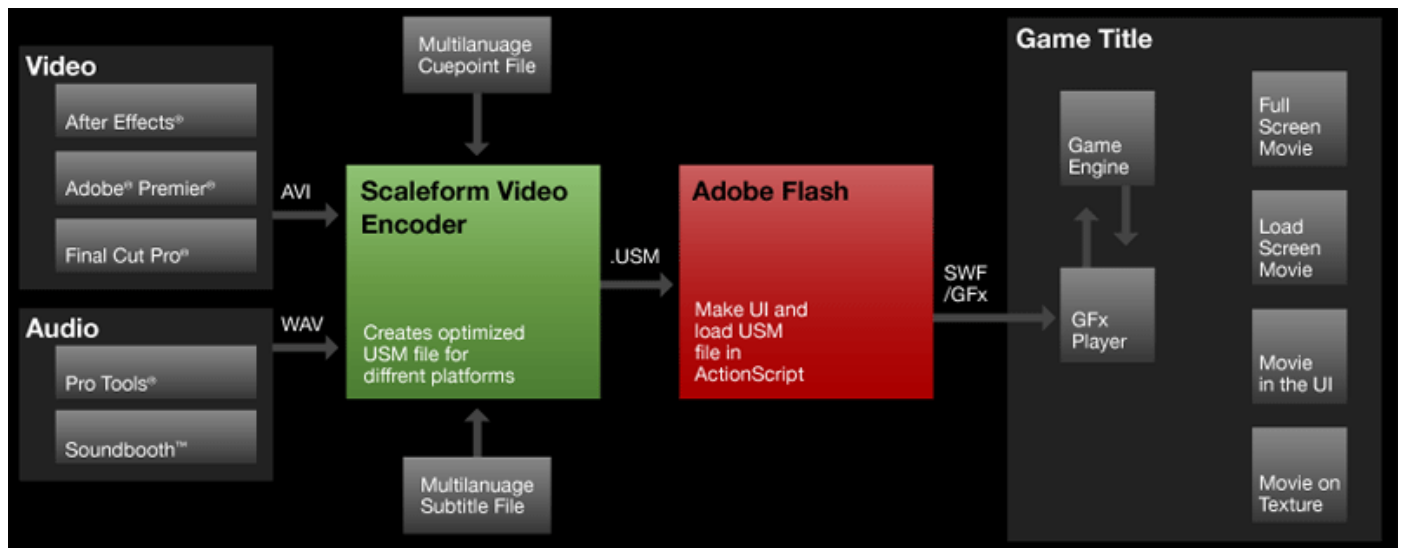
Scaleform

Scaleform

USM

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video\scaleform_logo.usm

2.8



1 Scaleform

1. Scaleform API.
2. Adobe Premiere AVI
3. Scaleform AVI USM
4. Adobe Flash SWF USM
5. ActionScript
6. SWF
7. SWF

* USM Flash ActionScript

Flash

3 Scaleform

Scaleform™ 4.3

CRI

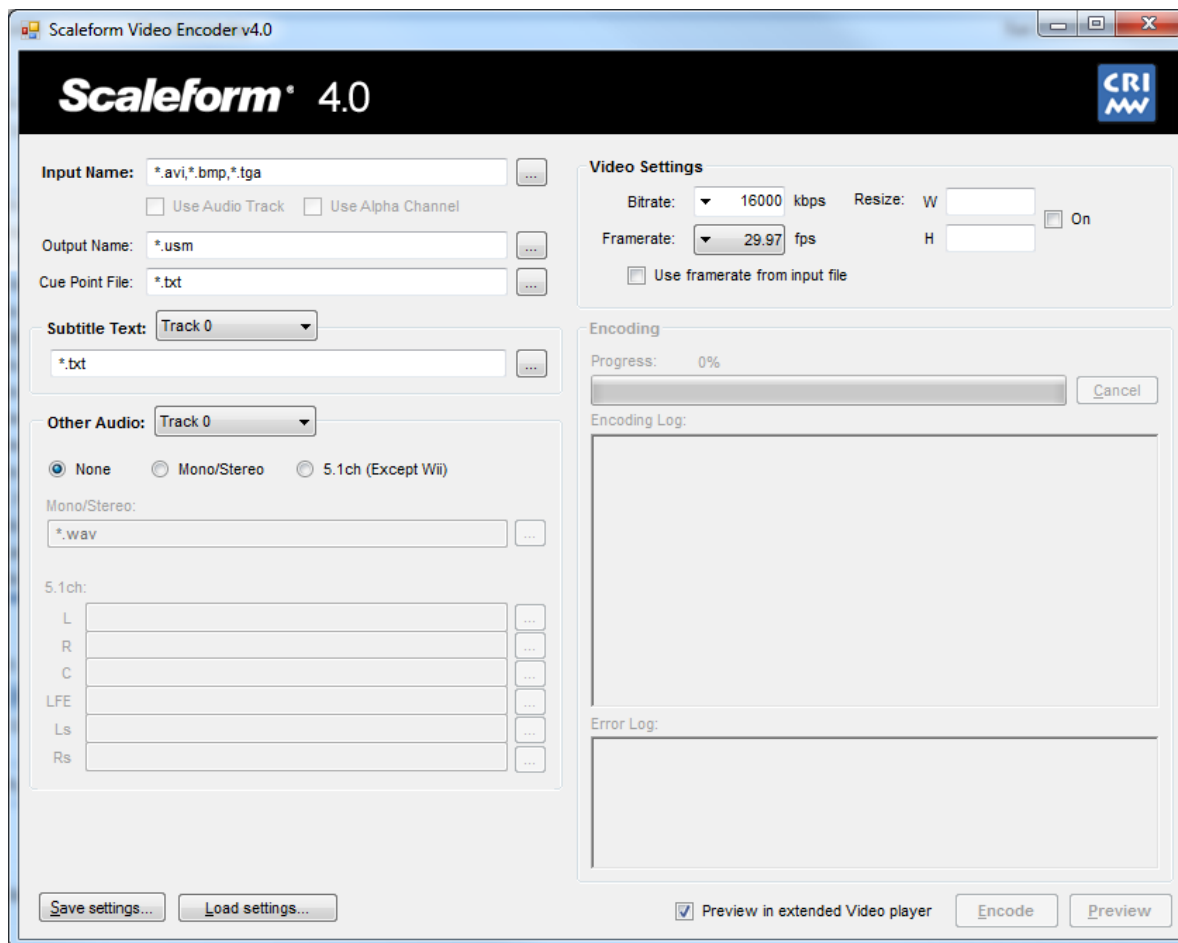
USM

CRI

Scaleform

Scaleform

USM



2: Scaleform

3.1

USM

USM

alpha

2

Windows™

Scaleform

Scaleform

Windows Vista: *Start->All Programs->Scaleform->GFX SDK 4.3->Video*

Windows XP: *Start->Programs->Scaleform->GFX SDK 4.3->Video*

1. Browse  AVI
- a. AVI BMP TGA
- b. BMP TGA

Input Name:

☐ Use Audio Track ☐ Use Alpha Channel

Output Name:

3:

-
- ```

graph TD
 S1[1. Click 'Encode' button] --> S2[2. Select audio format
BMP TGA AVI
Use Audio Track]
 S2 --> S3[3. Select alpha channel
alpha USM
Use Alpha Channel]
 S3 --> S4[4. Select output name
USM browse]
 S4 --> S5[5. Click 'Encode' button]

```
2. Select audio format  
BMP TGA AVI  
Use Audio Track
3. Select alpha channel  
alpha USM  
Use Alpha Channel
4. Select output name  
USM browse
5. Click 'Encode' button

## 3.2

- 1.
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3. *Encode to*
4. *Preview* *USM*

### 3.3

SWF

USM

Cue Point File

Browse

Cue Point File:

4:

10000

5000

5

—  
— 5000/1000=5

DVD

=

Flash

[ActionScript](#)

time, cue point type (0 or 1), cue point name, parameter1, parameter2, ...  
parameter10

:

1000, 0, cue\_point\_1, my\_param=5, my\_other\_param=10

#### CuePoint.txt

```
1000
0,0,start
1000,0,cue1
2000,1,cue2,name1_0=value1_0,name1_1=value1_1
3000,0,cue3
4000,1,cue4,name4_0=value4_0
5000,0,cue5
6000,1,cue6,name6_0=value6_0,name6_1=value6_1
```

```

7000,0,cue7,name7_0=value7_0,name7_1=value7_1,name7_2=value7_2
8000,1,cue8,name8_0=value_0
9000,0,cue9
10000,1,cue10,name10_0=value10_0,name10_1=value10_1
11000,0,end

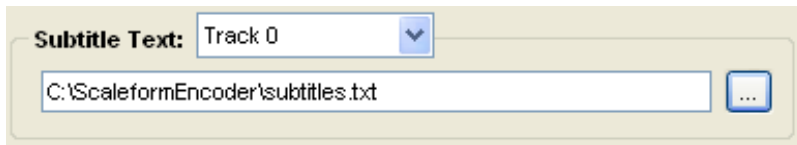
```

### 3.3.1

1.       Scaleform
  
2.       *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
  
3.       *Cue Point File*               *browse*               *sample\_cue\_points.txt*       —  
          *sample.avi*
  
4.       *Encode*
  
5.       *Preview*                       USM
  
6.       *fast forward*   *rewind*

## 3.4

1. *Subtitle* 32 —Tracks 0-31
2. *browse*



5:

1000

ID

ID

Flash

[ActionScript](#)

start time, end time, messageId

### Subtitle.txt

```
1000
2000, 5000, This is the first subtitle.
8000, 11000, subtitleId2
```

### 3.4.1

1. Scaleform
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3. Cue Point File *sample.avi* browse *sample\_cue\_points.txt* —
4. Subtitle Text *sample.avi* browse *sample\_subtitles.txt* —
5. Encode
6. *getting\_started\_with\_video\_tutorial.swf* Scaleform Player Scaleform
7. 2-5 8-11

3.5

/ 5.1

1. Scaleform Other Audio
2. — / 5.1 Wii  
5.1 Wii 5.1
3. / Mono/Stereo Browse WAV
4. Use Audio Track

Other Audio: Track 0

☒ None

☐ Mono/Stereo

☐ 5.1ch (Except Wii)

Mono/Stereo:

\*.wav

5.1ch:

L

R

C

LFE

Ls

Rs

6:

5. 5.1 Browse — L R C LFE Ls Rs —

5.1  
L —  
R —  
C — ( )  
LFE —

**Ls –**

**Rs –**

### 3.5.1

1. Scaleform
2. *C:\Program Files\Scaleform\GFx SDK 4.3 \Bin\Tools\VideoEncoder\sample.avi*
3. Cue Point File *sample.avi* browse *sample\_cue\_points.txt* —
4. Subtitle Text *sample.avi* browse *sample\_subtitles.txt* —
5. Other Audio ‘Track 1 ’
6. Mono/Stereo
7. Mono/Stereo browse *sample\_audio.wav*
8. Encode
9. Preview USM
10. Audio Channel SubtitleAudio Channel

3.6

Bitrate

Video Settings

Bitrate: 16000 kbps

Resize: W

☐ On

Framerate: 29.97 fps

H

☐ Use framerate from input file

7:

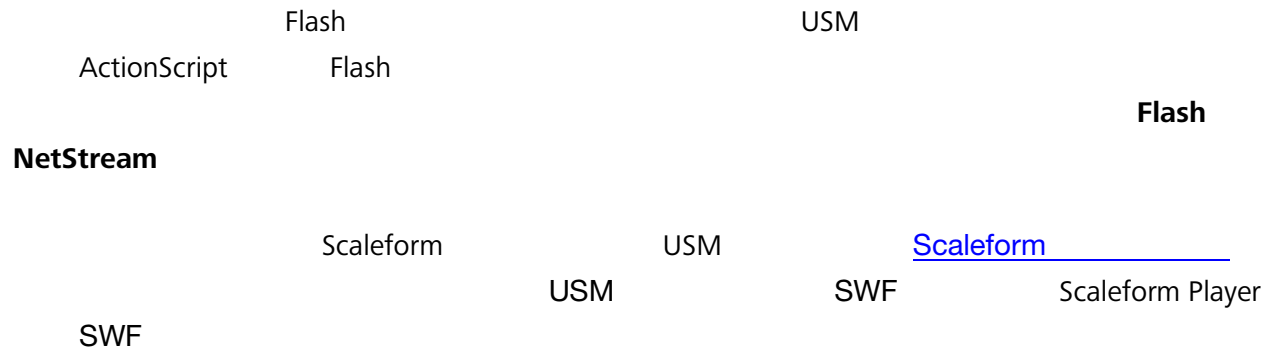
1. Bitrate —

| (kbps) |       |         |      |       |
|--------|-------|---------|------|-------|
|        | PS3   | Xbox360 | Wii  | PC    |
| 1080p  | 40000 | 40000   | n/a  | 40000 |
| 1080p  | 30000 | 30000   | n/a  | 30000 |
| 1080p  | 20000 | 20000   | n/a  | 20000 |
| 720p   | 36000 | 36000   | n/a  | 36000 |
| 720p   | 26000 | 26000   | n/a  | 26000 |
| 720p   | 16000 | 16000   | n/a  | 16000 |
| 480p   | 16000 | 16000   | 8000 | 16000 |
| 480p   | 12000 | 12000   | 6000 | 12000 |
| 480p   | 8000  | 8000    | 4000 | 8000  |

2. Framerate 29.97 fps— NTSC AVI  
Use framerate from input file

- PC 60 / NTSC 29.97 / 60Hz
3. *Resize W H W H*  
*On*
4. *Encode Preview*  
 USM
5. *Preview*  
*Encode Preview in extended Video player option*

## 4 Flash



## 4.1 Flash

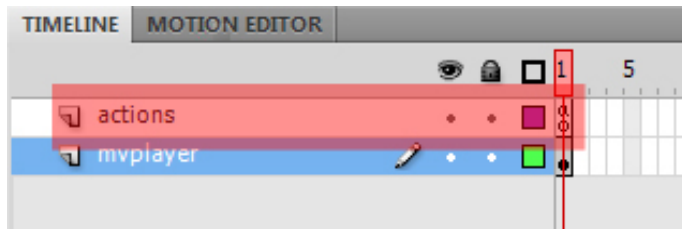
AS 2.0 Flash  
SWF

Scaleform

USM

1. *Layer 1* mvplayer

2. *New Layer* 'action'



8: mvplayer

3. *actions* F9 *Actions* *Actions*

```
_global.gfxExtensions = true;
_focusrect = false;
onLoad = function()
{
 mvplayer.playVideo("sample.usm");
}
```

SWF

USM

"C:/pathtovideo/sample.usm"

4. *mvplayer*

5.

a.

b.

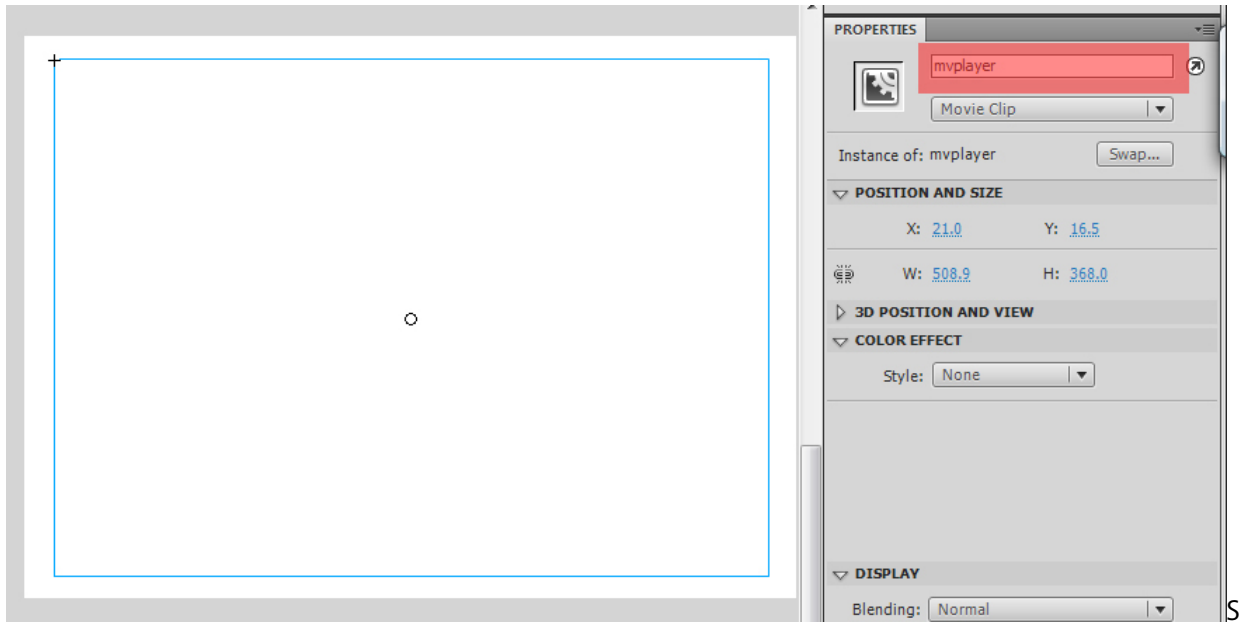
c.

d.

*Convert to Symbol*

'mvplayer'

OK



## 9: mvplayer

6. new mvplayer

7. Properties ' mvplayer '

8. mvplayer

9. mvplayer ' actions '

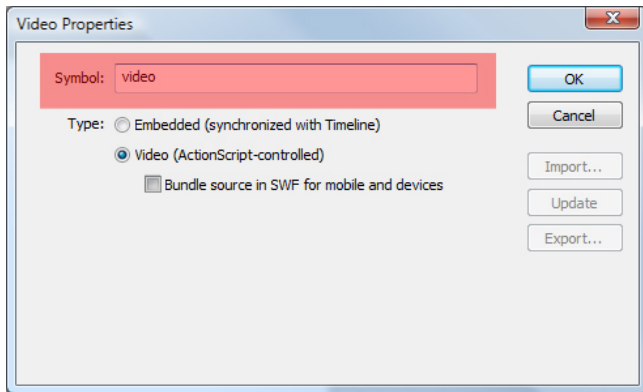
10. actions 1 Actions

```
nc = new NetConnection();
nc.connect(null);
ns = new NetStream(nc);
video.attachVideo(ns);
this.attachAudio(ns);
var audio_sound:Sound = new Sound(this);
function playVideo(videoToPlay:String):Void
{
 ns.play(videoToPlay);
}
```

mvplayer actions 1 Layer 1

11. Library New Video

12. ' Video' OK



**10:**

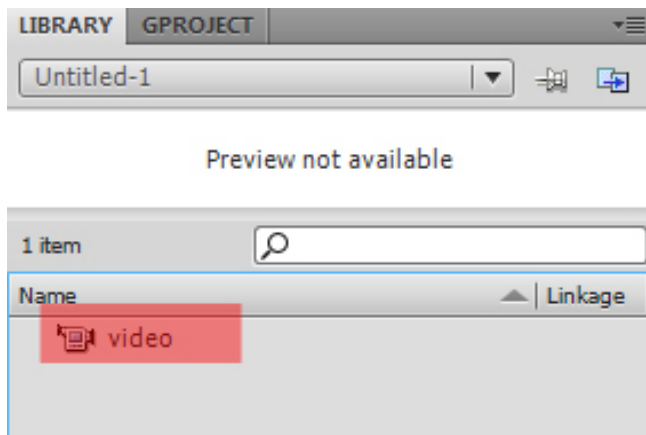
13. *New Layer* button

‘ Layer 3 ’

14. *Library*

*Layer 3*

*Library*



**11: Library**

15. *Properties*

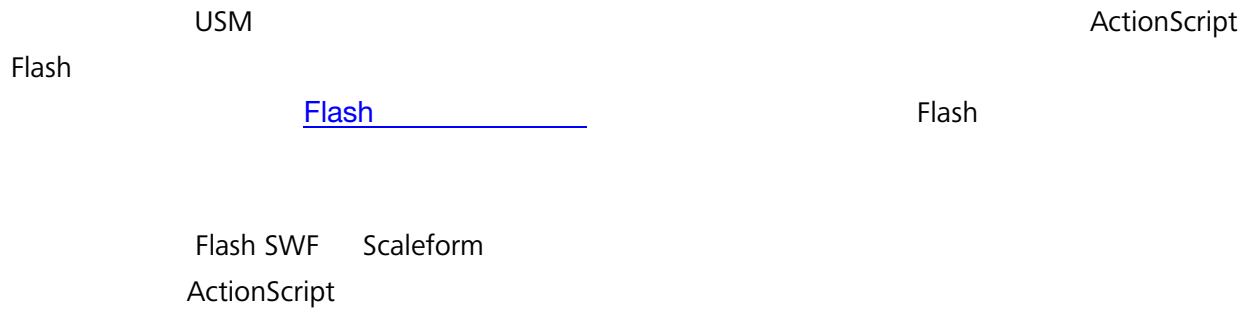
‘ Video ’

16.

## 4.2

- |    |       |                  |                  |
|----|-------|------------------|------------------|
|    |       | SWF              | Scaleform Player |
| 1. | Flash | Scaleform        | SWF              |
| 1. |       | Scaleform Player |                  |
| 2. | SWF   | Scaleform Player |                  |
|    | a.    | SWF              | Scaleform Player |

## 5 ActionScript



## 5.1 ActionScript

USM

/ Flash

Flash SWF / USM

USM

Flash

ActionScript

*mvplayer*

*actions* 1

1. *mvplayer*

2. *mvplayer* *actions* *Actions*

3. *traceCuePoint*

*ns.onCuePoint* = *traceCuePoint*;

4. *traceCuePoint* *metaPropPJParams*

(*currentCuePoint.name*) (*currentCuePoint.time*)

(*currentCuePoint.type*) *ns.time*

```
function traceCuePoint(currentCuePoint:Object):Void
{
 var metaPropPJParams:Object = currentCuePoint.parameters;
 trace("\t\t name: " + currentCuePoint.name);
 trace("\t\t time: " + currentCuePoint.time);
 trace("\t\t type: " + currentCuePoint.type);
 if (metaPropPJParams != undefined)
 {
 trace("\t\t parameters:");
 traceObject(metaPropPJParams, 4);
 }
}
```

5. *traceObject*

SWF

```

function traceObject(obj:Object, indent:Number):Void
{
 var indentString:String = "";
 for (var j:Number = 0; j < indent; j++)
 {
 indentString += "\t";
 }
 for (var i:String in obj)
 {
 if (typeof(obj[i]) == "object")
 {
 trace(indentString + " " + i + ": [Object]");
 traceObject(obj[i], indent + 1);
 }
 else
 {
 trace(indentString + " " + i + ": " + obj[i]);
 }
 }
}

```

|    |                  |                  |           |     |
|----|------------------|------------------|-----------|-----|
| 6. | Flash            | Scaleform Player | Scaleform | SWF |
|    | Scaleform Player |                  |           |     |
|    | Scaleform Player |                  |           |     |

## 5.2 ActionScript

|                   |                  |                                                                           |   |              |                |     |                     |       |  |
|-------------------|------------------|---------------------------------------------------------------------------|---|--------------|----------------|-----|---------------------|-------|--|
|                   | Flash            |                                                                           |   |              |                |     |                     |       |  |
|                   |                  | Flash SWF                                                                 | / |              |                | USM |                     |       |  |
| <i>sample.usm</i> |                  | <u>Flash</u>                                                              |   | <u>Flash</u> |                |     |                     |       |  |
| 1.                | <i>mvplayer</i>  |                                                                           |   |              |                |     |                     |       |  |
| 2.                | <i>mvplayer</i>  | <i>actions</i>                                                            |   |              | <i>Actions</i> |     |                     |       |  |
| 3.                |                  | subtitleChannelNumber                                                     |   |              |                | 0   |                     |       |  |
|                   |                  | <code>var subtitleChannelNumber = 0;</code>                               |   |              |                |     |                     |       |  |
| 4.                |                  |                                                                           |   |              |                | 0   |                     |       |  |
|                   |                  | <code>ns.onMetaData = function(infoObject:Object)</code>                  |   |              |                |     |                     |       |  |
|                   |                  | <code>{</code>                                                            |   |              |                |     |                     |       |  |
|                   |                  | <code>    ns.subtitleTrack = 1;</code>                                    |   |              |                |     |                     |       |  |
|                   |                  | <code>    subtitleChannelNumber = infoObject.subtitleTracksNumber;</code> |   |              |                |     |                     |       |  |
|                   |                  | <code>    trace("Subtitle Channels: " + subtitleChannelNumber);</code>    |   |              |                |     |                     |       |  |
|                   |                  | <code>}</code>                                                            |   |              |                |     |                     |       |  |
| 5.                |                  | <i>mvplayer</i>                                                           |   |              |                |     | <i>' subtitle '</i> |       |  |
| 6.                |                  |                                                                           |   |              |                | USM |                     | Msg   |  |
|                   |                  | subtitleId2                                                               |   |              | ActionScript   |     |                     | Msg   |  |
|                   | 5                | <i>subtitle dynamic</i>                                                   |   |              |                |     |                     |       |  |
|                   |                  | <code>ns.onSubtitle = function(msg:String)</code>                         |   |              |                |     |                     |       |  |
|                   |                  | <code>{</code>                                                            |   |              |                |     |                     |       |  |
|                   |                  | <code>    trace("Subtitle: " + msg);</code>                               |   |              |                |     |                     |       |  |
|                   |                  | <code>    subtitle.text = msg;</code>                                     |   |              |                |     |                     |       |  |
|                   |                  | <code>}</code>                                                            |   |              |                |     |                     |       |  |
| 7.                | Flash            | Scaleform Player                                                          |   | Scaleform    |                | SWF |                     |       |  |
|                   | Scaleform Player |                                                                           |   |              |                |     |                     | Flash |  |

## 5.3 ActionScript

USM

SWF

Flash SWF

/

USM

Flash

*mvplayer*

*actions*

1

*Actions*

1. *mvplayer*

2. *mvplayer* *actions*

*Actions*

3. AudioTracks

CurAudioTrack

0

```
var AudioTracks:Array;
var CurAudioTrack = 0;
```

4. audioTracks

audioTracks

```
 ' ns.onMetaData'
'AudioTracks = infoObject.audioTracks;' —
Channels: " + subtitleChannelNumber);'
ns.onMetaData = function(infoObject:Object)
{
 AudioTracks = infoObject.audioTracks;
}
 'trace("Subtitle
 '}'
```

5. *mvplayer*

‘prev\_audiotrack ’

6. ‘ mvplayer ’ ‘ action ’  
next\_audiotrack

*Actions*

```
prev_audiotrack.onRelease = function()
{
```

```

 if (AudioTracks == undefined || AudioTracks.length < 2)
 {
 return;
 }
 if (CurAudioTrack == 0)
 {
 CurAudioTrack = AudioTracks.length - 1;
 }
 else
 {
 CurAudioTrack--;
 }
 ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
 }

```

7.      *mvplayer*                              5                              'next\_audiotrack '

8.      ' mvplayer '      ' action '                              *Actions*  
          next\_audiotrack

```

next_audiotrack.onRelease = function()
{
 if (AudioTracks == undefined || AudioTracks.length < 2)
 {
 return;
 }
 CurAudioTrack++;
 if (CurAudioTrack >= AudioTracks.length)
 {
 CurAudioTrack = 0;
 }
 ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
}

```

9.      Scaleform Player                      Flash                              USM  
                                       *Next      Previous*

## 6 ActionScript

Scaleform    Flash            ActionScript  
                 ' NetStream '    Adobe

Scaleform  
NetStream

### 6.1            *NetStream*

currentFps:Number

```

 ns.onMetaData = function(infoObject:Object)
 {
 audioTrackArray = infoObject.audioTracks;
 }
 }

```

subtitleTracksNumber

USM

```

 ns.onMetaData = function(infoObject:Object)
 {
 numSubtitleChannels = infoObject.subtitleTracksNumber;
 trace("Subtitle Channels: " + numSubtitleChannels);
 }
 }

```

```
onStatus = function(infoObject:Object) {}
```

NetStream

Adobe Flash Help      ‘ onStatus ’

## 6.3      ***NetStream***

```
close()
```

ns.time      0

```
ns.close();
```

```
pause([flag:Boolean])
```

```
ns.pause(true);
```

```
play(name:Object, start:Number, len:Number, reset:Object)
```

USM

```

 ns.play("myVido.usm");

seek(offset:Number)

 ns.seek(50);

setBufferTime(bufferTime:Number)

 GFX::Video::Video
 (Bitrate)

 1

 ns.setBufferTime(2.5);

```

## 6.4 *Scaleform*

```

subtitleTrack = Number

 NetStream

 ns.subtitleTrack = 1;

audioTrack = Number

 NetStream

 ns.audioTrack = 2;

subAudioTrack = Number

 SubAudio NetStream

 ns.subAudioTrack = 3;

voiceTrack = Number

 5.1ch 5.1ch

 ns.voiceTrack = 1;

setReloadThresholdTime(reloadTime:Number)

```

|                   |   |                   |                    |   |
|-------------------|---|-------------------|--------------------|---|
|                   | 4 | setBufferTime     | (Reload Threshold) | 1 |
| Gfx::Video::Video | 4 |                   | 3                  |   |
|                   |   | Gfx::Video::Video |                    |   |

```

 ns.setReloadThresholdTime(1);

setNumberOfFramePools(numPools: Number)
 Gfx::Video::Video
 CPU 2
 ns.setNumberOfFramePools(3);

setOpenTimeout(timeout: Number)
 USM 5
 0
 ns.setOpenTimeout(5);

currentFrame: Number

 trace("Current FPS: " + ns.currentFps + " fps");

Loop = Boolean

 true NetStream false

 ns.loop = true;

clipRect = new rectangle(x: Number, y: Number, width: Number, height: Number)

 ns.clipRect = new rectangle(100, 100, 240, 200);

onSubtitle = function(msg: String) {}

 USM

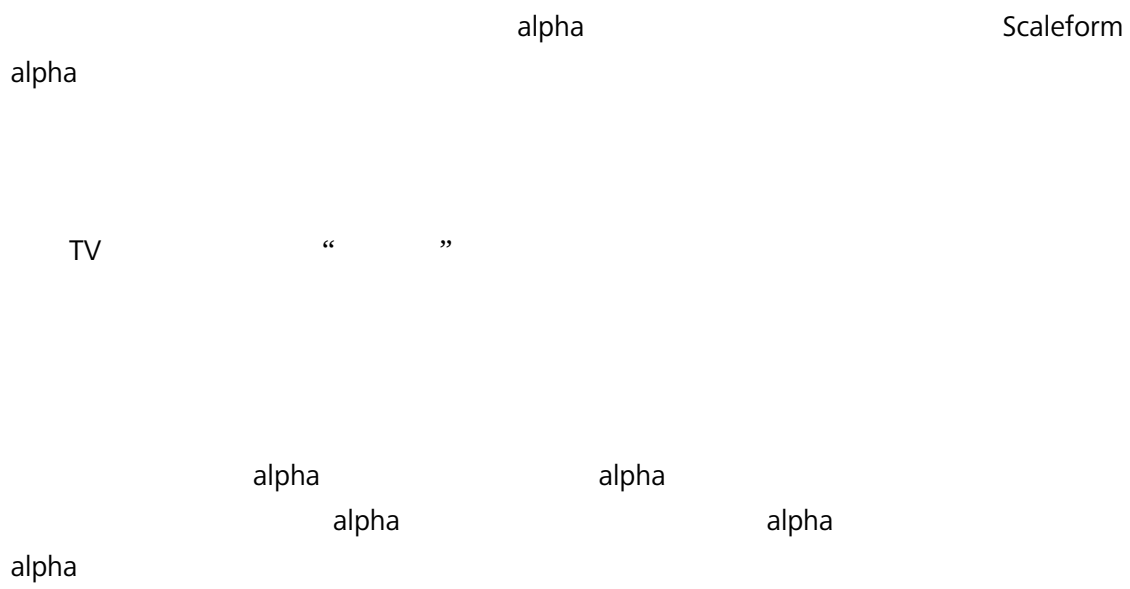
 ns.onSubtitle = function(msg: String) { trace(msg); }

```



/

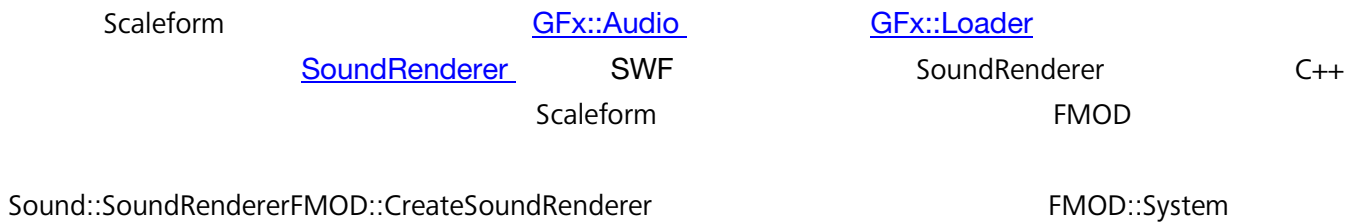
## 7.1 Alpha



## 8

Scaleform

### 8.1 Scaleform



```

//
FMOD::System* pFMOD = NULL;
result = FMOD::System_Create(&pFMOD);
if (result != FMOD_OK)
 exit(1);
result = pFMOD->getVersion(&version);
if (result != FMOD_OK || version < FMOD_VERSION)
 exit(1);
result = pFMOD->init(64, FMOD_INIT_NORMAL, 0);
if (result != FMOD_OK)
 exit(1);
Ptr<Sound::SoundRenderer> psoundRenderer =
 *SoundRendererFMOD::CreateSoundRenderer();
if (!psoundRenderer->Initialize(pFMOD))
 exit(1);

// Gfx::Audio
Ptr<Gfx::Audio> paudioState = *new Gfx::Audio(psoundRenderer);
loader.SetAudio(paudioState);

```

## 8.2

Scaleform

[Video](#)

Gfx::Loader

[VideoSoundSystem](#)

Gfx::Video::Video

Scaleform

SoundRenderer

```
// GfxVideo
Ptr<Gfx::Video::Video> pvc = *new Video(Thread::NormalPriority);
// SoundRenderer instance which is based on
SoundRenderer

if (psoundRenderer)
{
 pvc->SetSoundSystem(psoundRenderer);
}
//
// SetSoundSystem Video
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new
 VideoSoundSystemDX8(0)));

loader.SetVideo(pvc);
```

Windows

Video::Video

VideoPC

affinity mask

```
VideoPC(Thread::ThreadPriority
 decodingThreadsPriority = Thread::NormalPriority,
 int decodingThreadsNumber = MAX_VIDEO_DECODING_THREADS,
 UInt32* affinityMask = NULL);

UInt32 affinities[] = {
 DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
 DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
 DEFAULT_VIDEO_DECODING_AFFINITY_MASK };

Ptr<Video::Video> pvc = *new Video::VideoPC(Thread::NormalPriority,
 MAX_VIDEO_DECODING_THREADS, affinities);
```

|     |         |           |       |
|-----|---------|-----------|-------|
| PS3 | Xbox360 | Scaleform | Video |
|-----|---------|-----------|-------|

|     |                      |   |
|-----|----------------------|---|
| PS3 | GfX::Video::VideoPS3 | 5 |
|-----|----------------------|---|

```
VideoPS3(CellSpurs* spurs,
 int ppu_num, Thread::ThreadPriority ppu_priority,
 int spu_num, UInt8* spurs_priorities = NULL);
```

|                    |           |
|--------------------|-----------|
| spurs -            | CellSpurs |
| spu_num -          | SPU       |
| ppu_num -          | PPU       |
| ppu_priority -     | PPU       |
| spurs_priorities - | SPU       |

```
UInt8 spurs_priorities[] = {1,1,1,1,0,0,0,0};
Ptr<Video::Video> pvc = *new VideoPS3(GetSpurs(), 0, 1000,
 4, spurs_priorities);
```

## PPU

|                 |     |           |      |     |
|-----------------|-----|-----------|------|-----|
| Scaleform Video | PPU | 50-60 FPS | 720p | PPU |
| 5%              |     |           |      |     |
| ,               |     |           |      |     |

|      |     |
|------|-----|
| PPU  |     |
| 40%  | PPU |
| 4-5% | 1%  |

```
UInt8 spurs_priors[] = {0,0,1,1,1,1,0,0};
Ptr<Video::Video> pVideo = *new VideoPS3(GetSpurs(), 0,
 Thread::BelowNormalPriority, 4, spurs_priors);
```

|         |              |
|---------|--------------|
| Xbox360 | VideoXbox360 |
|---------|--------------|

```
VideoXbox360(unsigned decodingProcs = Xbox360_Proc1 | Xbox360_Proc3 |
 Xbox360_Proc5, Thread::ThreadPriority
 decodingThreadsPriority = Thread::NormalPriority,
 Xbox360Proc readerProc = Xbox360_Proc2);
```

```
Ptr<Video::Video> pvc = *new VideoXbox360(Xbox360_Proc1 |
 Xbox360_Proc3|Xbox360_Proc5,
 Thread::NormalPriority,
 Xbox360_Proc2);
```

## 8.3

## API

Video::VideoBase::ReadCallback –

VideoBase::IsIORequired VideoBase::EnableIO

*ReadCallback*

```
//
class VideoReadCallback : public VideoBase::ReadCallback
{
 public:
 VideoReadCallback() {}
 virtual ~VideoReadCallback() {}

 virtual void OnReadRequested()
 {
 //
 VideoReadStart = Timer::GetTicks();
 // ...
 }
 virtual void OnReadCompleted()
 {
 //
 GameReadStart = Timer::GetTicks();
 // ...
 }

 UInt64 VideoReadStart;
 UInt64 GameReadStart;
};

//
Ptr<Video::Video> pvc = *new Video::VideoPC(
```

```
Thread::NormalPriority, MAX_VIDEO_DECODING_THREADS, affnts);
```

```
//
Ptr<VideoReadCallback> pvideoReadCallback = *new VideoReadCallback;
pvc->SetReadCallback(pvideoReadCallback);
```

*IsIORequired*

*EnableIO.*

*EnableIO*

/

```
if(pvc->IsIORequired())
{
 pDataLoader->Enable(false);
 pvc->EnableIO(true);
 fprintf(stderr, "Video is loading\n");
}

// ...

if(!pvc->IsIORequired())
{
 pvc->EnableIO(false);
 pDataLoader->Enable(true);
 fprintf(stderr, "Game data is loading\n");
}
```

VideoBGLoadFlash    VideoBGLoadData

*Apps\Samples\Video*

- VideoBGLoadFlash – Flash
- VideoBGLoadData –

## 8.4 Scaleform VideoSoundSystem

[GFX::Video::VideoSoundSystem](#)

Video::Video playback

[Video::SetSoundSystem\(\)](#)

Video

- Video::VideoSoundSystemDX8 – Windows DirectSound
- Video::VideoSoundSystemXA2 – Windows Xbox360 XAudio2
- Video::VideoSoundSystemPS3 – MultiStream for PS3
- Video::VideoSoundSystemWii – Wii
- Video::VideoSoundSystemFMOD - FMOD
- Video::VideoSoundSystemWwise – Wwise

Scaleform

Flash

Video::VideoSoundSystem

Scaleform

Sound::SoundRenderer

8.1

VideoSoundSystem

VideoSound

SoundRenderer

SoundRenderer

Video

```
#include "Video/Video_VideoSoundSystemXA2.h"
```

```
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));
```

[GFX::Video::VideoSoundSystem](#)

[GFX::Video::VideoSound](#)

GFX::VideoSoundSystem

GFX::VideoSoundSystem

VideoSound

Create

Scaleform

VideoSound

Scaleform

VideoSound::PCMStream

VideoSoundSystem

VideoSoundSystemWwise

Wwise SDK

v2009.2.1 build 3271. Audiokinetic

Visual Studio /

SDK\Samples\Plugins\AkAudioInput. Wwise Gfx::Video::Video Wwise  
 SDK

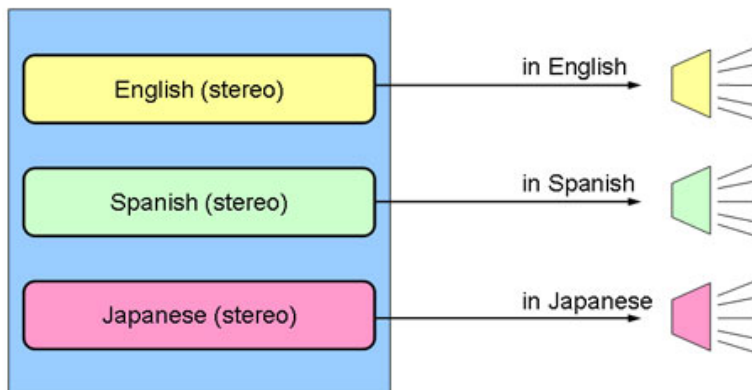
```
#include "Video\Video_VideoSoundSystemWwise.h"
Ptr<Video::VideoSoundSystem> wwise =
 Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemWwise());
pvc->SetSoundSystem(wwise);
wwise->Update();
```

Scaleform Player - FxPlayerAppBase.cpp FxSoundFMOD.cpp FxSoundWwise.cpp  
 Apps\Samples\FxPlayer Apps\Samples\Common  
 Video\_VideoSoundSystem\*.cpp Src\Video

## 8.5

Gfx::Video::Video ActionScript  
 NetStream *audioTrack*

```
ns.audioTrack = 2;
Multilingual Movie Data
```



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5.1

Gfx::Video::Video

- 1.
2. SubAudio

## 8.5.1

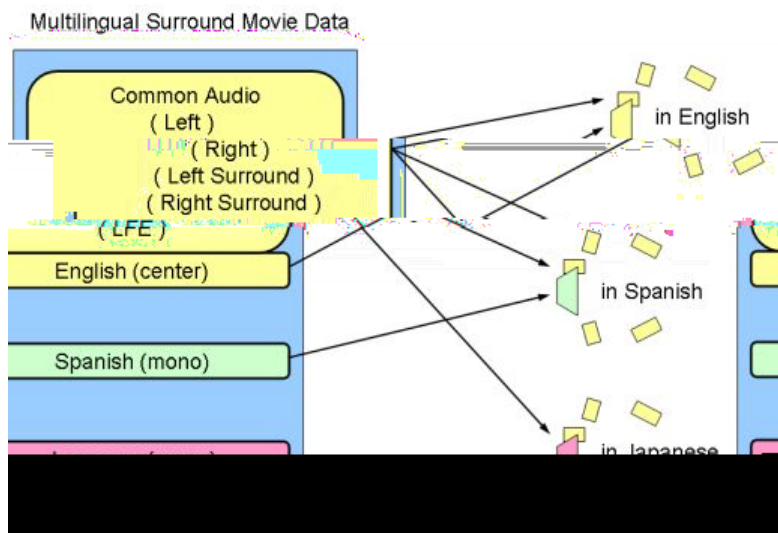
5.1

5.1

5.1

```
ns.audioTrack = 0; // main 5.1 audio track
ns.voiceTrack = 5; // mono audio track
```

5.1



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## 8.5.2 SubAudio

SubAudio

GfX::Video::Video

GfX::Video::Video

SubAudio

```
ns.audioTrack = 0; // main audio track
ns.subAudioTrack = 6; // secondary audio track
```

SubAudio Playback

Sound.setVolume()

```
var audio:Sound = new Sound(this);
audio.setVolume(80, 50); // set volume for main and subaudio tracks
```

## 14 SubAudio

## 9

(Stuttering)

1. IO

2.

3.

1. - 720P 3 4  
8

2. - /  
Scaleform

3. - ActionScript

6

- a. `NetStream.setNumberOfFramePools(numPools:Number)`
- b. `NetStream.setBufferTime(bufferTime:Number)`